



ASSIGNMENT 1

Cryptography

MAPHOKA MAHLOMOLA
224028650

Reflective Report

Part (d)

I. Collision Resistance

Replacing a single letter in the transaction text, such as replacing “Transaction1” with “transaction1” yielded entirely different SHA-256 hashes during the practical exercise. This is a demonstration of the avalanche effect, whereby even slight changes in the input yield entirely different hashes. Furthermore, SHA-256 is a collision resistant algorithm, which means that it is not possible to find two inputs that yield the same hash. The ability to be collision resistant is necessary in a financial ledger since if two transactions could yield the same hash, then someone would be able to substitute a valid transaction for a forged transaction without affecting the hash.

II. Pre-Image Resistance

Pre-image resistance is a property whereby, given a hash value, it is computationally infeasible to determine the original message that produced it. Financial systems can, therefore, store hash values without disclosing transaction information. In the Proof of Work exercise, it showed that increasing the difficulty from three to four leading hexadecimal zeros led to a substantial increase in the number of iterations and execution time.

III. What Hashing does not provide

For Part (a), when hashing the same input, the SHA-256 digest will always be the same, confirming that hashing is deterministic. Hashing does not provide secrecy because there is no secret key. This means the same input will produce the same digest. Authentication and non-repudiation cannot be achieved using hashing because the hash does not provide proof of the person who performed the action. In addition, hashing cannot ensure data availability or control who has access to the data in the ledger.

IV. Conclusion

The combination of collision resistance and pre-image resistance makes SHA-256 a powerful algorithm to ensure data integrity in financial ledgers. The practical exercise has shown that even a slight change in input will result in a totally new hash value, and raising the Proof-of-Work difficulty will require much more computing power. Nevertheless, hashing is only a part of the overall security process in a financial environment. It is also necessary to have digital signatures, encryption, and proper access control measures.

Github link: <https://github.com/Maphoka-M/Assignment1-Cryptography>